

MICROECONOMIC ASPECTS OF ECONOMIC DEVELOPMENT

Final Replication Paper

Table 1 to 8 of Chakravorty, Ujjayant; Dar, Manzoor H.; Emerick, Kyle (2019) : Inefficient water pricing and incentives for conservation, CESifo Working Paper, No. 7560, Center for Economic Studies and ifo Institute (CESifo), Munich

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Abstract

The authors conducted two RCTs (randomized controlled trials) in three districts: Mymensingh, Rangpur, and Rajshahi of Bangladesh to study a water conservation technology for rice production – AWD (Alternate Wetting and Drying). They have found out that this simple to implement and highly regarded water conservation scheme only saves water and increases profits in villages where farmers pay a marginal price for water, but not when they pay fixed seasonal charges; this is based on their first RCT. The second RCT tests the effectiveness of a policy intervention that encourages volumetric pricing and provides prepaid irrigation cards for hourly irrigation. This intervention favourably alters the demand for the water-saving technology; demand elasticity for AWD in the treatment group fell by 33%, purchase of AWD went up by 35% , and, consumer surplus at the median price increased by over 50%. This highlights the role of pricing mechanisms in agricultural water use and shows that policy intervention can have significant positive effects on the demand for water-conservation technology. The findings of the authors are relevant especially for developing countries where the price of water is fixed, despite an increase in water scarcity.

Background & Introduction

One challenge many developing countries face is food production and procurement of food for its population; majority of which is dependant on the rainfed farms. Over the years, Many developing countries have successfully increased food production by irrigating their crops in the dry season, through technologies that use water more efficiently such as – Alternate Wetting and Drying (AWD). Alternate Wetting and Drying (AWD) is a water-saving technology that farmers can apply to reduce their irrigation water consumption in rice fields without decreasing its yield; a theoretically perfect solution. In AWD, irrigation water is applied a few days after the disappearance of the ponded water; hence, the field gets alternately flooded and non-flooded. [2] In this paper, the authors conduct two randomized controlled trials to study how fixing the market failure of zero marginal price for water affects the usage, impact, and demand for AWD. The results show that introducing a marginal price can increase the demand for such water-saving technology.

The authors conduct this study in 3 districts of Bangladesh. They argue that the reason why AWD technology is not widely adopted is due to a market failure, in the form of a zero marginal price for water. The farmers, in the districts surveyed by the author, do not pay for the amount of water they use – rather a flat fee based on the area of land they cultivate is charged; indicative of no incentive present for farmers to use water-saving technologies, such as AWD (there is no cost savings associated with reducing their water use). [1] The 2 RCTs reveal that the intervention led to a significant increase in the usage of AWD technology among farmers, indicating that a zero marginal price of water can be a major barrier to the adoption of water-saving technologies. The authors also suggest that fixing this market failure by introducing a marginal price for water can change the demand for water-saving technology and result in significant water savings in agriculture.

Methodology

The study was conducted in three districts of Bangladesh— Mymensingh, Rangpur, and Rajshahi. The water pricing varied considerably due to differences in groundwater table lengths or pricing mechanisms; 12 upazilas were identified in these districts, each consisting of approximately 260 villages. The authors selected a sample of villages by randomly drawing 400 villages split across the districts; it was ensured by the field staff that visited each village that the farmers were growing rice during the boro season. If not, the village was replaced with another randomly drawn village from the same district. Once approved by the field staff, the field teams worked with a village leader to identify ten farmers cultivating land near the village tube well; for each of these farmers, the plot located closest to the tube well was mapped out and referred to as the "study plot".

Subsequently, the 4,000 farmers (10 farmers from 400 villages) were visited for a baseline survey in November-December of 2016 to collect information on household demographics, agricultural production, water management, and water prices for the study plot and one other randomly selected plot of each farmer. Before the start of boro cultivation in 2017, the authors randomly assigned each village to one of two groups. In the first group, the field staff trained the ten farmers on the purpose of alternate wetting and drying (AWD) and how to use it. In contrast, nothing was done in the remaining 200 villages, which served as the control. The authors then went on to have a follow up survey in July 2017 after the boro rice crop had been harvested. This survey collected information on self-reported irrigation management, input use, crop yield, revenue, and profit.

Replication Results & Summary

Table 1

Table 1: Effects of AWD Treatment on Water Usage

	(1)	(2)	(3)	(4)
	Water Level	Dry Field	Water Level	Dry Field
AWD Treatment	-0.0614 (0.161)	0.0191 (0.0228)	0.119 (0.220)	-0.0122 (0.0265)
AWD Treatment *			-0.544*	0.0960*
Volumetric Pricing			(0.287)	(0.0495)
			.	.
Volumetric Pricing			-0.107 (0.333)	-0.0583 (0.0605)
Strata Fixed Effects	Yes	Yes	Yes	Yes
p-Value:			0.021	0.047
Treat+Treat*Volumetric				
Control Mean	2.32	0.45	2.32	0.45
Number of Obs.	7598	7598	7596	7596
R squared	0.033	0.035	0.036	0.037

Standard errors are given in parenthesis.

The data are from random unannounced visits to the study plots of sample farmers during the 2017 boro (dry) growing season. The dependent variable in columns 1 and 3 is the amount of standing water in the field, measured in centimeters. The dependent variable in columns 2 and 4 is an indicator variable for a dry field with no standing water. Volumetric pricing is an indicator for farmers for whom the water price is tied to usage, either through hourly charges or fuel payments. Standard errors are clustered at the village level. Asterisks indicate that coefficient is statistically significant at the 1% ***, 5% **, and 10% * levels.

The first column shows the effect of AWD treatment on the amount of standing water in the field, measured in centimeters. The coefficient of (1) is negative which shows that AWD treatment reduces the amount of standing water in the field in villages where farmers pay a marginal price for water. AWD treatment does not have a significant effect on the amount of standing water in villages. However, The average effect of AWD on water management — across the entire sample — is both small and statistically insignificant. AWD is only effective for farmers who face **volumetric**

water prices. In column 3, AWD generates an effect on water levels only for farmers facing nonzero marginal prices. Introducing AWD in places with volumetric pricing lowers the amount of observed irrigation water by 0.43 centimeters, or an 18 percent decrease. The probability of a plot being dry also increases by 19%. Finally, the third row of the table shows that the correlation between volumetric pricing and water use is small and statistically insignificant.

Table 2

Table 2: Separate effects by time of growing season

	0-70 Days		70+ Days	
	(1)	(2)	(3)	(4)
	Water Level	Dry Field	Water Level	Dry Field
Treatment	-0.350** (0.152)	0.0590** (0.0265)	0.250 (0.286)	-0.0211 (0.0334)
Strata Fixed Effects	Yes	Yes	Yes	Yes
Control Mean	2.706	0.325	1.857	0.594
Number of Obs.	4188.00	4188.00	3410.00	3410.00
R squared	0.020	0.035	0.085	0.113

Standard errors are given in parenthesis.

The period refers to after planting, i.e, 0-70 after planting; 70+ days after planting. The data are from random unannounced visits to the study plots of sample farmers during the 2017 boro (dry) growing season. Columns 1 and 2 are for measurements taken up to 70 days after transplanting. Columns 3 and 4 are for measurements taken more than 70 days after transplanting. The dependent variable in columns 1 and 3 is the amount of standing water in the field, measured in centimeters. The dependent variable in columns 2 and 4 is an indicator variable for a dry field with no standing water. Volumetric pricing is an indicator for farmers for whom the water price is tied to usage, either through hourly charges or payments for diesel fuel. Standard errors are clustered at the village level. Asterisks indicate that coefficient is statistically significant levels. 1% ***, 5% **, and 10% * levels.

Table 2 shows that treatment effects exist only during the first 70 days of the growing season; the results for which are statistically significant at the 99% confidence interval with p-value less than 0.01. Treatment plots had about 35 percent less water and were about 5 percent more likely to be dry during the first 70 days after planting. In contrast, we do not see a statistically significant difference between treatment and control plots after 70 days.

Note: The authors pre-specified this split in the data to approximately divide the season into the time before and after the start of flowering. Farmers practice AWD during the time up to flowering.

Table 3

Table 4: Effects of AWD treatment on water usage

	0-70 Days		70+ Days	
	(1)	(2)	(3)	(4)
	Water Level	Dry Field	Water Level	Dry Field
AWD Treatment	-0.0485 (0.208)	-0.0120 (0.0325)	0.258 (0.376)	-0.00329 (0.0390)
AWD Treatment *	-0.788***	0.185***	0.0138	-0.0708
Volumetric Pricing	(0.287)	(0.0540)	(0.474)	(0.0746)
Volumetric Pricing	0.0256 (0.363)	-0.0824 (0.0650)	-0.488 (0.420)	0.0228 (0.0660)
Strata Fixed Effects	Yes	Yes	Yes	Yes
Control Mean	2.71	0.32	1.86	0.59
P Value	0.000	0.000	0.328	0.244
Number of Obs.	4187	4187	3409	3409
R squared	0.027	0.043	0.086	0.114

Standard errors are given in parenthesis.

The period refers to after planting, i.e, 0-70 after planting; 70+ days after planting. The data are from random unannounced visits to the study plots of sample farmers during the 2017 boro (dry) growing season. The dependent variable in columns 1 and 3 is the amount of standing water in the field, measured in centimeters. The dependent variable in columns 2 and 4 is an indicator variable for a dry field with no standing water. Volumetric pricing is an indicator for farmers for whom the water price is tied to usage, either through hourly charges or fuel payments. Standard errors are clustered at the village level.

The first two columns refer to the period of 0-70 days after planting; the last two columns refer to 70+ days after planting. The coefficient estimates for the AWD Treatment variable in columns 1 and 2 are negative and statistically insignificant – indicating that the treatment did not have a significant effect on the water levels of the districts. The coefficient estimate for the interaction term is negative and statistically significant in column 1, indicating that the AWD treatment was more effective in reducing water usage for farmers who had a volumetric pricing system.

Table 4

Table 6: Table 4: Effects on self-reported water use

	Number Irrigations		Times Drained	
	(1)	(2)	(3)	(4)
Treatment	-3.589*** (0.486)	-3.590*** (0.607)	2.207*** (0.225)	1.888*** (0.258)
Treatment *		-0.015		0.918*
Volumetric Pricing		(0.994)		(0.497)
Volumetric Pricing		1.082 (1.263)		0.032 (0.433)
Upazila Fixed Effects	Yes	Yes	Yes	Yes
Mean in Control	19.10	19.10	2.42	2.42
p-Value: Treat+Treat*Volumetric		0.000		0.000
Number of Observations	3985	3984	3983	3982
R squared	0.539	0.540	0.359	0.366

Standard errors are given in parenthesis.

The data are taken from the followup survey after harvesting. The dependent variables are the number of times the field was irrigated (columns 1-2) and the number of times the field was drained or dried (columns 3-4). Standard errors are clustered at the village level. Asterisks indicate that coefficient is statistically significant levels. 1% ***, 5% **, and 10% * levels.

The table shows the effects of the treatment on the number of irrigations and the number of times the field was drained; with and without the interaction of the treatment with volumetric pricing, respectively. The AWD treatment has a statistically significant negative effect on the number of irrigations accross all columns; indicating that the treatment reduces the amount of water used for irrigation. Similarly it increases the frequency of draining or drying the field. The negative coefficient of the interaction term in the second column suggests that the effect of the treatment on the number of irrigations is moderated by volumetric pricing; which is not statistically significant. The positive coefficient of the interaction term in the fourth column suggests that the effect of the treatment on the number of times the field was drained or dried is moderated by volumetric pricing; which is statistically significant at the 5% level.

Table 8: Table 5: Effects of AWD on costs, revenues, and profits

Log								
	(1) Profit	(2) Water Cost	(3) Yield	(4) Revenue	(5) Profit	(6) Water Cost	(7) Yield	(8) Revenue
T	-338.316 (901.831)	133.089 (124.857)	4.051 (30.189)	178.885 (820.028)	-0.036 (0.046)	0.023 (0.024)	-0.001 (0.014)	0.001 (0.017)
T *	2205.179* (1293.913)	-435.469 (279.224)	10.295 (37.221)	1222.695 (1153.819)	0.122** (0.061)	-0.081 (0.053)	0.009 (0.017)	0.029 (0.022)
VP								
VP	-711.475 (1332.595)	372.471 (226.509)	27.800 (37.246)	350.272 (1321.963)	-0.123* (0.070)	0.066 (0.043)	0.013 (0.018)	0.003 (0.028)
Strata FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
P-Value:	0.049	0.226	0.515	0.091	0.035	0.225	0.417	0.045
Observations	3982	3983	3982	3982	3932	3983	3982	3982
R squared	0.298	0.365	0.352	0.390	0.273	0.347	0.329	0.351

Standard errors are given in parenthesis.

T stands for AWD Treatment, VP stands for Volumetric Pricing.

The data are taken from the followup survey after harvesting. The dependent variables are profit per acre (column 1), water cost in taka per acre (column 2), crop yield in kilograms per acre (column 3), and revenue in taka per acre (column 4). Columns 5 through 8 show the same regressions with the log of profit, water cost, yield, and revenue, respectively. Standard errors are clustered at the village level. Asterisks indicate that coefficient is statistically significant at the 1% ***, 5% **, and 10% * levels.

Table 5

The table presents the regression results of the effects of AWD on costs, revenues, and profits. The table has four dependent variables – profit per acre, water cost in taka per acre, crop yield in kilograms per acre, and revenue in taka per acre.

AWD treatment reduces profit per acre by -338.316 taka; which is statistically insignificant, it also increases water cost by 133.089 taka per acre, but this coefficient is not statistically significant as well. AWD Treatment has no significant effect on crop yield whilst, reducing revenue per acre by 178.885 taka. Upon the interaction of the AWD treatment and Volumetric Pricing, we can see that there exists a positive and significant effect on the profit (log of profit per acre as well). At the same time, it has a negative and significant effect on the log of profit per acre for those farmers who have water available to them at volumetric rates.

Table 6

Table 10: Table 6: Effects separately by card ownership in villages with prepaid irrigation pumps

	(1) Water Cost	(2) Profit	(3) Log Cost	(4) Water Cost	(5) Profit	(6) Log Cost
T	108.3 (358.0)	1210.6 (1202.1)	0.0260 (0.0438)	404.4 (786.4)	1668.9 (1980.1)	0.0678 (0.0671)
T * HC	-1039.4** (485.1)	2524.1 (2074.7)	0.112 (0.0773)	-1177.4* (623.0)	2523.8 (2103.0)	0.100 (0.0714)
HC	1184.3*** (409.1)	-1253.7 (1872.9)	-0.0722 (0.0708)	1100.7** (484.4)	-1895.2 (1687.2)	-0.0629 (0.0583)
Strata Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Covariates	No	No	No	Yes	Yes	Yes
Treatment*Covariates	No	No	No	Yes	Yes	Yes
Mean in Control	5611.93	29999.22	10.27	5527.29	30473.33	10.29
p-Value: Treat+Treat*Has Card	0.006	0.028	0.030	0.104	0.023	0.005
Number of observations	1340	1340	1332	1276	1276	1270

Standard errors are given in parenthesis.

T stands for Treatment and HC stands for Has Card

The data are from the follow up survey and are limited to the Rajshahi district where some farmers have their own prepaid irrigation card to pay for water by the hour. The variable “Has Card” is an indicator variable for farmers that report having their own prepaid card. The dependent variables are the cost of water per acre (columns 1 and 4), profit per acre (columns 2 and 5), and log profit per acre (columns 3 and 6). Columns 4-6 include demeaned farmer covariates from baseline and interactions between these demeaned covariates and the AWD treatment indicator. The covariates included are all of those in Table A1 (age, years of education, household size, number of livestock owned, landholdings, television ownership, refrigerator ownership, tube well ownership, indicator for knowledge of AWD, indicator for a rented or sharecropped plot, plot area, number of crops grown, indicator for growing two rice crops, number of boro irrigations, revenue per acre in boro, boro total cost per acre, and aman revenue per acre). Asterisks indicate that coefficient is statistically significant at the 1% ***, 5% **, and 10% * levels.

This table shows the results of regressions examining the effects of AWD technology separately for farmers who have their own prepaid irrigation card (HC) and those who do not. The dependent variables are the cost of water per acre, profit per acre, and log profit per acre. The first three columns of the table correspond to farmers who do not have their own prepaid card; the last three columns corresponding to farmers who do have their own prepaid card.

The treatment group or e with access AWD technology had to paid on average 108.3 more than the control group; however, they earned on average 1210.6 taka more than the control group as well. The interaction coefficient shows whether the treatment effect differs between farmers who have their own prepaid card and those who do not. It can be seen that the water cost for those farmers who have a card is lower that those with it by 1039.4 taka. This shows that having a prepaid irrigation card to pay for water benfits the farmers to a greater extend.

Table 7

Table 12: Table 7: Impacts of volumetric pricing treatment on demand for water-saving technology

	(1)	(2)	(3)	(4)
Volumetric Treatment	0.0430 (0.0436)	-0.1428 (0.1044)	0.0353 (0.0428)	-0.5510** (0.2622)
AWD Price	-0.0105*** (0.0008)	-0.0129*** (0.0012)		
AWD Price * Volumetric Treatment		0.0034** (0.0015)		
Log AWD Price			-0.5084*** (0.0351)	-0.6123*** (0.0489)
Log AWD Price* Volumetric Treatment				0.1497** (0.0654)
Strata Fixed Effects	Yes	Yes	Yes	Yes
Mean in Control	0.413	0.413	0.413	0.413
Elasticity at Price=55 Treat	-1.26	-1.14	-1.25	-1.13
Elasticity at Price=55 Control	-1.39	-1.70	-1.37	-1.70
P-value: Equal Elasticities		0.009		0.025
Number Obs	3569	3569	3569	3569
R squared	0.249	0.254	0.256	0.260

Standard errors are given in parenthesis.

The data are from the 144 villages that were part of the second-year experiment. The sample consists of 25 farmers per village. The dependent variable in all regressions is an indicator if the farmer purchased the AWD pipe at the randomly set price. Prices were set randomly at the village level and range from 20 to 90 taka (around 0.24 to 1.1). The volumetric treatment variable is an indicator for villages where the 25 farmers were provided assistance with filling out the application for a prepaid (hourly) irrigation card and a waiver of the 150 taka sign-up fee. The p-value for equal elasticities is based on standard errors from the delta method. Standard errors are clustered at the village level. Asterisks indicate that coefficient is statistically significant at the 1% ***, 5% **, and 10% * levels.

The coefficient for the volumetric treatment variable is 0.0430 and is statistically significant at the 10% level; this implies that volumetric pricing treatment has a positive effect on the demand for water-saving technology.

Looking at the interaction between the volumetric treatment variable and the AWD price, we can see that the coefficient is 0.0034 and is statistically significant at the 5% level. This indicates that the effect of the volumetric pricing treatment on the demand for water-saving technology is moderated by the AWD price. This is an interesting observation by the authors.

We can also see the demand for water-saving technology decreases as the AWD price increases and that the effect of the volumetric pricing treatment on the demand for water-saving technology is moderated by the AWD price.

Table 8

Table 14: Table 8: Impacts of volumetric pricing treatment on installation of water-saving technology

	(1)	(2)	(3)	(4)
Volumetric Treatment	0.0200 (0.0278)	-0.1071 (0.1074)	0.0187 (0.0279)	-0.4848 (0.3321)
AWD Price	-0.0016*** (0.0006)	-0.0033** (0.0014)		
AWD Price * Volumetric Treatment		0.0023 (0.0015)		
Log AWD Price			-0.0763** (0.0307)	-0.1665** (0.0739)
Log AWD Price * Volumetric Treatment				0.1287 (0.0795)
Strata Fixed Effects	Yes	Yes	Yes	Yes
Mean in Control	0.068	0.068	0.068	0.068
Elasticity at Price=55 Treat	-1.01	-0.60	-0.95	-0.45
Elasticity at Price=55 Control	-1.31	-2.58	-1.23	-3.08
P-value: Equal Elasticities		0.001		0.005
Number Obs	3600	3600	3600	3600
R squared	0.033	0.041	0.033	0.043

Standard errors are given in parenthesis.

The data are from the 144 villages that were part of the second-year experiment. The sample consists of 25 farmers per village. The dependent variable in all regressions is an indicator equal to one if it was verified that the farmer installed AWD on one of their plots. Prices were set randomly at the village level and range from 20 to 90 taka (around 0.24 to 1.1). The volumetric treatment variable is an indicator for villages where the 25 farmers were provided assistance with filling out the application for a prepaid (hourly) irrigation card and a waiver of the 150 taka sign-up fee. Standard errors are clustered at the village level. The p-value for equal elasticities is based on standard errors from the delta method. Asterisks indicate that coefficient is statistically significant at the 1% ***, 5% **, and 10% *levels.

The coefficient of the volumetric treatment variable is 0.0200, which indicates that the treatment increased the likelihood of installation by 2 percentage points. However, this coefficient is not statistically significant. Similarly column 2 indicates that higher AWD prices reduce the likelihood of installation. The interaction term between the AWD price and the volumetric treatment variable is positive and statistically significant, suggesting that the effect of AWD prices on the likelihood of installation depends on whether the village received the volumetric pricing treatment or not.

About the Replication Exercise

The authors have done a one of a kind, comprehensive study on a simple and easy to implement water conservation technology AWD (Alternate wetting and drying) with tremendous future potential, specially with rising water scarcity. They looked at it from the lens of water pricing, such as the zero marginal cost of water; there seems to be a lack of incentive for the farmers to shift to such water conservation technologies; however, the author does not stop at that goes one step further to examine what types of interventions will have a greater benefit in terms of profits to the farmers.

The replication package, however, despite being organised is a mess. None of the tables in the replication package correspond to tables in the paper. The regressions do generate the same output as used in the paper.

Conclusion & Discussion

Overall, we can see from these tables that the volumetric pricing treatment may be an effective policy tool for promoting the adoption of water-saving technology among farmers in Bangladesh, and a similar model could be used in other developing countries as well ; its effectiveness, however, may depend on several factors, including access to information, income level, plot size, access to prepaid irrigation and baseline water use. Alternative policies and awareness programs, specially in countries like India, may be needed to address the needs of farmers who are not responsive to changes in water prices or who face structural barriers to adopting water saving technology, in this case AWD.

This paper is specially relevant in today's times as many regions of the world are facing water scarcity, and this problem is expected to worsen due to climate change, population growth, and urbanization. The authors provide empirical evidence for the need for water pricing reform as a means to increase water conservation. After having come across this paper, I am able to draw connections between the authors results from Bangladesh, droughts in India and the climate change crisis. The yearly temperature averages have been increasing constantly; due to which, states like Maharashtra have already started to witness an increase in droughts and a reduction in rainfall. In such times of water scarcity, it is the need of the hour to switch to water conservation technologies such as AWD.

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